KOR-ACC-83 Routing Protocol Flap SOP for Huawei Access Node

Node: KOR-ACC-83 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_8480

Description

The Routing Protocol Flap (BGP route churn, OSPF adjacency loss, ISIS LSP flooding) failure in the Access domain of Huawei's Node KOR-ACC-83 at the Koramangala Datacenter is primarily caused by a complex interplay of factors, including but not limited to, misconfigured or incompatible routing policies, suboptimal network design, and resource-related issues such as memory or CPU constraints. The underlying protocol mechanics, particularly in relation to BGP, OSPF, and ISIS, may also be compromised due to factors like routing table overflow, adjacency instability, or LSP flooding. Furthermore, hardware-related issues, such as faulty interfaces or switching silicon, can also contribute to this failure domain. In this SOP, we will delve into the diagnostic and resolution procedures for addressing this critical failure, strictly adhering to the Huawei CLI dialect.

Pre-requisite Checks

- Verify that all redundant links are operational and functioning as expected.
- Ensure that the Koramangala Datacenter is accessible and all necessary personnel are on-site.
- Confirm that the Huawei Node KOR-ACC-83 is powered on and operational.

Potential Root Causes

Cause	Probability
Misconfigured or incompatible routing policies	High
Suboptimal network design	Medium
Resource-related issues (memory or CPU constraints)	Low
Faulty interfaces or switching silicon	Medium

Diagnostic Commands & Expected Outputs

```
Command: display bgp process, Expected Output: BGP Process Information (Process ID 1), Description: This command will display detailed information about the BGP process, including the current state, routing table statistics, and any errors or warnings.

Command: display ospf interface, Expected Output: OSPF Interface Information (Interface ID 1), Description: This command will display detailed information about the OSPF interface, including the current state, neighbor information, and any errors or warnings.

Command: display isis database, Expected Output: ISIS Database Information (Database ID 1), Description: This command will display
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detailed information about the ISIS database, including the current
state, LSP information, and any errors or warnings.
```

Resolution Steps

Step 1: Step: Verify and correct routing policy configurations, Command: display bgp policy, Description: Verify that all routing policy configurations are correct and consistent across all routers in the network. If any discrepancies are found, correct the policy configurations accordingly.

Step 2: Step: Optimize network design, Command: display network topology, Description: Verify that the network design is optimal and free from any bottlenecks or suboptimal paths. Make adjustments as necessary to ensure efficient routing and minimal congestion.

Step 3: Step: Address resource-related issues, Command: display system resource, Description: Verify that system resources, such as memory and CPU, are within acceptable limits. If any issues are found, address them accordingly to ensure the system is functioning within normal parameters.

Step 4: Step: Replace faulty interfaces or switching silicon, Command: replace interface [interface ID], Description: If a faulty interface or switching silicon is identified, replace it with a new one to ensure the system is functioning correctly.

Post-Check Verification

```
Command: display bgp process, Expected Output: BGP Process Information
(Process ID 1), Description: Verify that the BGP process is
functioning correctly and that any previous issues have been resolved.

Command: display ospf interface, Expected Output: OSPF Interface
Information (Interface ID 1), Description: Verify that the OSPF
interface is functioning correctly and that any previous issues have
been resolved.

Command: display isis database, Expected Output: ISIS Database
Information (Database ID 1), Description: Verify that the ISIS
database is functioning correctly and that any previous issues have
been resolved.
```

Roll-back Procedures

- Step: Revert changes made to routing policy configurations, Command: undo bgp policy, Description: If any changes were made to the routing policy configurations during the resolution process, revert them to their original state.

- Step: Restore original network design, Command: restore network topology, Description: If any changes were made to the network design during the resolution process, restore the original design.

- Step: Restore original system resource settings, Command: restore system resource, Description: If any changes were made to the system resource settings during the resolution process, restore the original settings.

- Step: Replace faulty interfaces or switching silicon with original components, Command: replace interface [interface ID] with original component, Description: If a faulty interface or switching silicon was replaced during the resolution process, replace it with the original component.